Horizontal Gene Transfer: stochastic VS deterministic models

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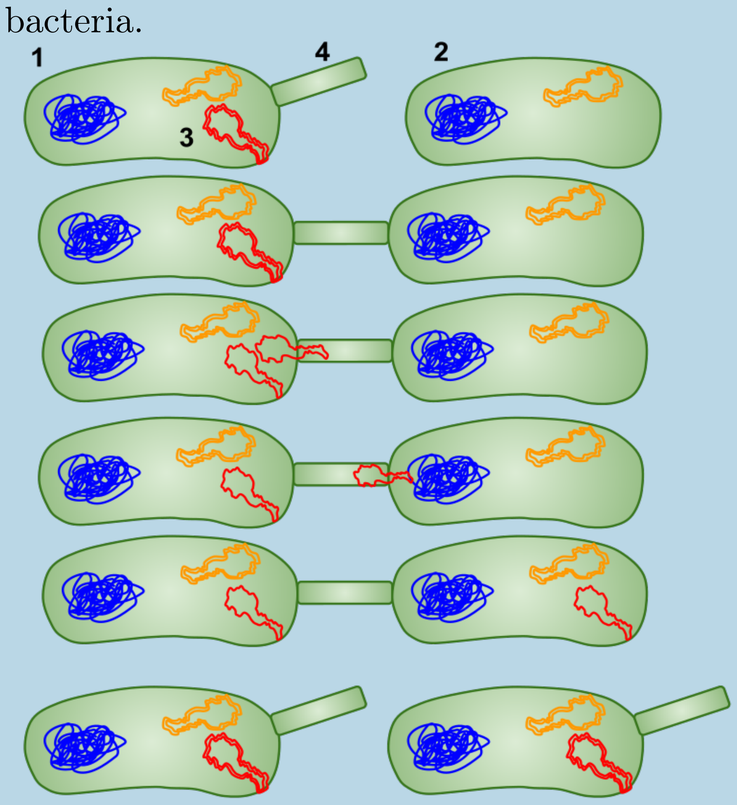
Problem

Horizontal Gene Transfer (HGT) — exchange of genetic material between unicellular/multicellular organisms not through a vertical transmission and plays a fundamental role in the evolution of organisms, the transmission of pathogens and the resistance to antibiotics in bacteria.



Notations and definitions

The demography of the population:

- $X_i(t)$: **trait** of i-th individual at **time** t
- K: **scale parameter** (*carrying capacity*), N^t — size of the population at time t
- **Birth:** $b(x)$ with a mutation kernel $\mathbf{M(x,y)}$.
- **Death + Competition:** $d(x) + C\frac{N^t}{K}$,
- **Horizontal Transfer (HGT):**

$$h_K(x,y,N^t) = \tau\alpha(x-y)\frac{K}{N^t}$$

α: (e.g: **heaviside**, **arctan**,...)
τ: transfer rate
C: competition rate

Project Structure

Stochastic model

$K \to +\infty$

PDE

$\frac{t}{\varepsilon} \to \infty \quad \varepsilon \to 0$

Hamilton-Jacobi PDE

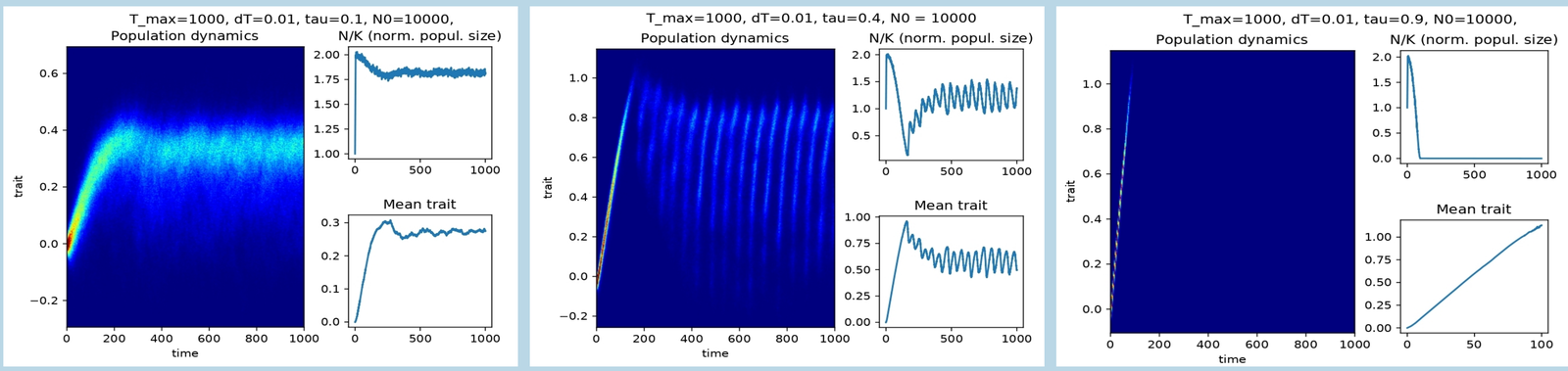
Asympt. Preserv. Scheme

$$\begin{array}{ccc} P_\varepsilon & \xrightarrow{\varepsilon\to 0} & P_0 \\ \uparrow{\scriptstyle h\to 0} & & \uparrow{\scriptstyle h\to 0} \\ S^h_\varepsilon & \xrightarrow{\varepsilon\to 0} & S^h_0 \end{array}$$

Individual Based Model

For a population described by point measure $\nu^t(dx) = \frac{1}{K}\sum_{i=0}^{N^t}\delta_{X_i(t)}(dx)$ the generator of the process is:

$$\begin{aligned} L^K F(\nu) &= \sum_{i=1}^N b(x_i)\int_{\mathbb{R}}\left(F\left(\nu+\tfrac{1}{K}\delta_y\right)-F(\nu)\right)M(x_i,dy) \\ &+ \sum_{i=1}^N \left(d(x_i)+C\tfrac{N}{K}\right)\left(F\left(\nu-\tfrac{1}{K}\delta_{x_i}\right)-F(\nu)\right) \\ &+ \sum_{i,j=1}^N h_K(x_i,x_j,\nu)\left(F\left(\nu+\tfrac{1}{K}\delta_{x_i}-\tfrac{1}{K}\delta_{x_j}\right)-F(\nu)\right). \end{aligned}$$

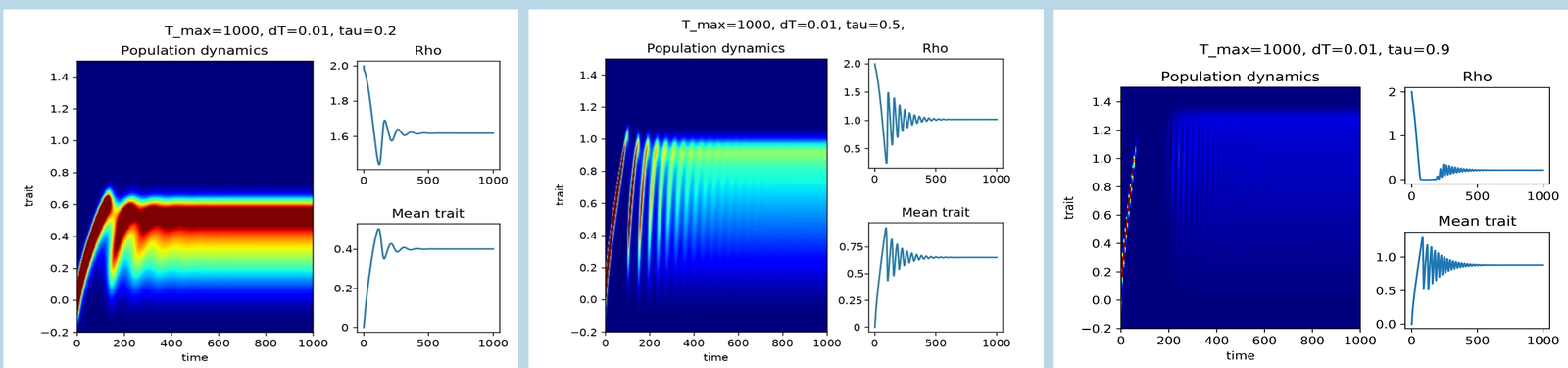


Integro-Differential Model

The sequence of stochastic processes gives a macroscopic approximation when $K \to +\infty$.

$$\begin{cases} \partial_t f(t,x) = -(d(x)+C\rho(t))f(t,x) + \displaystyle\int_{\mathbb{R}} M(x-y)b(y)f(t,y)dy + f(t,x)\int_{\mathbb{R}}\tau(x-y)\frac{f(t,y)}{\rho(t)}dy, \\ \rho(t) = \displaystyle\int_{\mathbb{R}} f(t,x)dx, \qquad f(0,x) = f^0(x) > 0. \end{cases}$$

- $f(t,x) := \lim_{K\to+\infty}\nu^K$: density of the population at time t with trait x,
- $\tau(y-x) := \tau_0\left[\alpha(x-y)-\alpha(y-x)\right]$: HT kernel

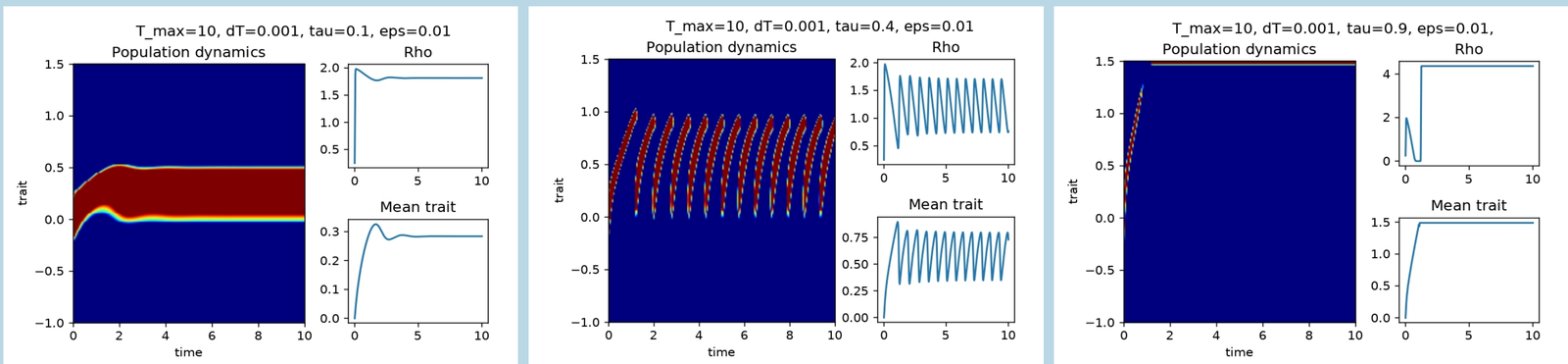


Hamilton-Jacobi equation

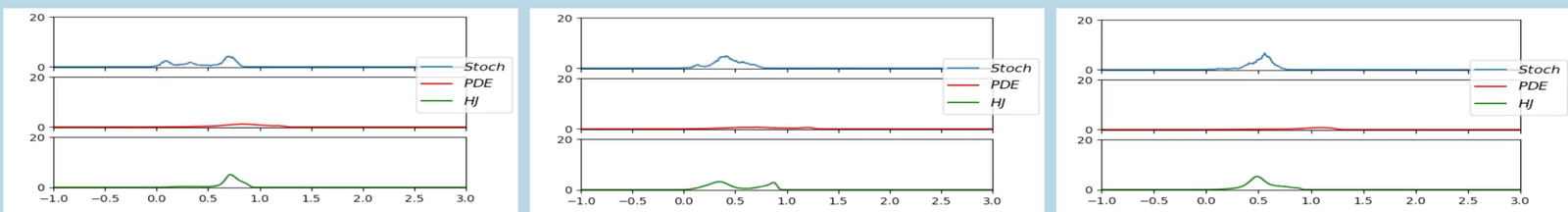
- Hopf-Cole transformation $\boxed{f_\varepsilon(t,x) = e^{\frac{u_\varepsilon(t,x)}{\varepsilon}}}$ where $u_\varepsilon \longrightarrow u$ as $\varepsilon \to 0$ *viscosity* solution of:

$$\partial_t u = -(d(x)+C\bar\rho(t)) + b(x)\int_{\mathbb{R}} M(z)e^{z\cdot\nabla_x u}dz + \tau(x-\bar x(t)), \qquad \text{with } \sup_{x\in\mathbb{R}} u(t,\cdot) \le 0.$$

- Population concentrates on $\bar x(t) = \operatorname{argmax} u(t,\cdot) \Leftrightarrow \boxed{f_\varepsilon(t,\cdot) \rightharpoonup \bar\rho(t)\delta(\cdot-\bar x(t))}$ as $\varepsilon \to 0$
- To catch Dirac concentration phenomenon: $\varepsilon \to 0 \Rightarrow$ Asymptotic Preserving (AP) scheme



Real Time Three Models Comparison



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